

Tableau Data Visualization Operations on Adult Dataset

Aim

To perform different data visualization operations using Tableau on the Adult Dataset.

a. 1D (Linear) Data Visualization

Visualization Used

Histogram

Tableau Section	Field
Columns	Age
Rows	Number of Records / adult_dataset.csv (Count)
Chart Type	Histogram

Purpose

To visualize the distribution of Age.

b. 2D (Planar) Data Visualization

Visualization Used

Scatter Plot

Tableau Section	Field
Columns	Age
Rows	Hours-Per-Week
Color (Optional)	Income
Chart Type	Scatter Plot

Purpose

To analyze the relationship between Age and working hours.

c. 3D (Volumetric) Data Visualization

Visualization Used

Bubble Chart

Tableau Section	Field
Columns	Age
Rows	Hours-Per-Week
Size	Capital-Gain
Color	Income
Detail	Education
Chart Type	Bubble Chart

Purpose

To represent three variables simultaneously using position and bubble size.

d. Temporal Data Visualization

Visualization Used

Line Chart

Tableau Section	Field
Columns	Age
Rows	AVG(Hours-Per-Week)
Chart Type	Line Chart

Purpose

To observe trends in working hours across age.

e. Multidimensional Data Visualization

Visualization Used

Heat Map

Tableau Section	Field
Columns	Education
Rows	Occupation
Color	Hours-Per-Week
Size	Age
Chart Type	Heat Map

Purpose

To analyze multiple attributes together.

f. Tree / Hierarchical Data Visualization

Visualization Used

Treemap

Tableau Section	Field
Label/Text	Occupation
Size	Hours-Per-Week
Color	Income
Chart Type	Treemap

Purpose

To represent hierarchical categorical data using rectangles.

g. Network Data Visualization

Visualization Used

Packed Bubble Chart

Tableau Section	Field
Detail	Occupation
Color	Education
Size	Hours-Per-Week
Chart Type	Packed Bubble Chart

Purpose

To represent relationships among categories.

Conclusion

The Tableau visualizations performed on the Adult dataset helped analyze distributions, relationships, trends, hierarchical structures, and multidimensional patterns using various graphical techniques.